Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/07/2004
PI Name:	Pavalko , Fredrick M Ph.D.		
Project Title:	Regulation Of Osteoblast Apoptosis By Fluid Shear Stress		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	fpavalko@iupui.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	317-274-3140
Organization Name:	Indiana University-Purdue University - Indianapolis		
PI Address 1:	Department of Cellular and Integrative Physiology		
PI Address 2:	Medical Science Bldg., Room 2069		
PI Web Page:			
City:	Indianapolis	**State:**	IN
Zip Code:	46202-5120	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1606		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Bone mass is increased in response to mechanical loading, while chronic unloading of bone, such as occurs during prolonged bed rest and in microgravity during space flight, results in atrophy of bone. Exposure of osteoblasts and osteocytes to fluid shear stress (FSS), induced by the movement of interstitial fluid within the spaces inside bone, has been hypothesized to play a role in the response of bone to mechanical loading. This hypothesis is supported by in vitro studies demonstrating that bone-forming osteoblasts and osteocytes are stimulated by FSS. Among the responses of bone cells to FSS are the release of autocrine/paracrine signaling factors including prostaglandins and nitric oxide (NO), both of which are known to play a role in bone growth and remodeling. In this project we are seeking to gain a better understanding of the cellular and molecular events that mediate the response of bone to loading. We are examining the effects of steady and pulsatile FSS on growth and survival of osteoblasts. Our results to date suggest that FSS promotes osteoblast survival in vitro. These results raise the possibility that mechanical loading may promote bone formation by increasing the lifespan of osteoblasts in bone through inhibition of osteoblast apoptosis. Our primary objective is to test		

	the hypothesis that FSS plays a role in regulating osteoblast survival using an in vitro cell culture model. Identify cellular mechanisms that regulate apoptosis in bone cells. Inhibition of osteoblast apoptosis could improve bone growth and remodelling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have continued to investigate the effects of fluid shear stress (FSS) on osteoblast apoptosis using three osteoblast cell types: primary rat calvarial osteoblasts, MC3T3-E1 osteoblastic cells and UMR106 osteosarcoma cells. We have followed up the results of our initial study which was published in 2003 (Pavalko, F.M., Gerard, R.L., Ponik, S.M., Gallagher, P.J., Jin, Y., and Norvell, S.M. (2003) Fluid shear stress inhibits TNF-alpha-induced apoptosis in osteoblasts: a role for shear stress-induced activation of PI3-kinase and inhibition of caspase-3. Journal of Cellular Physiology 194: 194-205). Cells were treated with TNF-alpha in the presence of cyclohexamide to rapidly induce apoptosis. Osteoblasts showed significant signs of apoptosis within 4-6 hours of exposure to TNF-alpha and cyclohexamide, and application of FSS (12 dynes/cm2) significantly attenuated this TNF-alpha-induced apoptosis. FSS activated PI3-kinase signaling, induced phosphorylation of Akt, and inhibited TNF-alpha-induced activation of caspase-3. Inhibition of PI3-kinase, using LY294002, blocked the ability of FSS to rescue osteoblasts from TNF-alpha-induced apoptosis and blocked FSS-induced inhibition of caspase-3 activation in osteoblasts treated with TNF-alpha. LY294002 did not, however, prevent FSS-induced phosphorylation of Akt, suggesting that activation of Akt alone is not sufficient to rescue cells from apoptosis. This result also suggests that FSS can activate Akt via a PI3-kinase-independent pathway. These studies demonstrate for the first time that application of FSS to osteoblasts in vitro results in inhibition of TNF-alpha-induced apoptosis through a mechanism involving activation of PI3-kinase signaling and inhibition of caspases. FSS-induced activation of PI3-kinase may promote cell survival through a mechanism that is distinct from the Akt-mediated survival pathway. We have also begun looking at the role of PKCzeta in fluid shear induced rescue of cells from apoptosis. Preliminary results suggest that an inhibitor of PKCzeta may block shear-induced osteoblast survival. This study should be submitted for publication by the end of 2004.
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Norvell SM, Ponik SM, Bowen DK, Gerard R, Pavalko FM. "Fluid shear stress induction of COX-2 protein and prostaglandin release in cultured MC3T3-E1 osteoblasts does not require intact microfilaments or microtubules." J Appl Physiol. 2004 Mar;96(3):957-66. Epub 2003 Nov 14. PMID: 14617531 , Mar-2004
Articles in Peer-reviewed Journals	Ponik SM, Pavalko FM. "Formation of focal adhesions on fibronectin promotes fluid shear stress induction of COX-2 and PGE2 release in MC3T3-E1 osteoblasts." J Appl Physiol. 2004 Jul;97(1):135-42. Epub 2004 Mar 5. PMID: 15004000 , Jul-2004
Articles in Peer-reviewed Journals	Shah R, Alvarez M, Jones DR, Torrungruang K, Watt AJ, Selvamurugan N, Partridge NC, Quinn CO, Pavalko FM, Rhodes SJ, Bidwell JP. "Nmp4/CIZ regulation of matrix metalloproteinase 13 (MMP-13) response to parathryoid hormone in osteoblasts." Am J Physiol Endocrinol Metab. 2004 Aug;287(2):E289-96. Epub 2004 Mar 16. PMID: 15026307 , Aug-2004
Articles in Peer-reviewed Journals	Norvell SM, Alvarez M, Bidwell JP, Pavalko FM. "Fluid shear stress induces ß-catenin signaling in osteoblasts." Calcified Tissue International, in press, October 2004. , Oct-2004